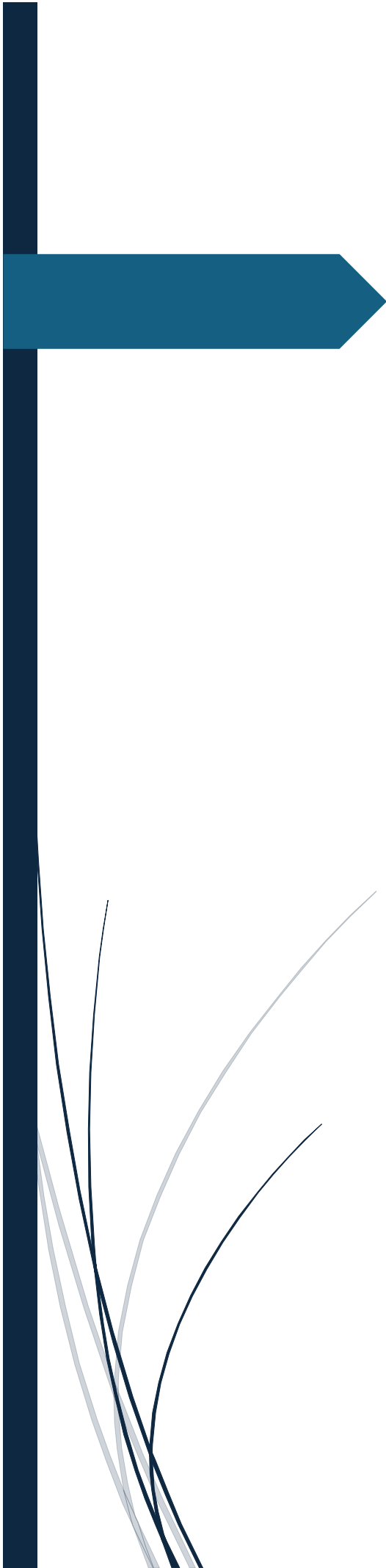




Happy Haven Insurance

NCD Schemes

Tanvir Randhawa
ACTL2102 ASSIGNMENT



Individual Assignment Cover Sheet

SUBMISSION DETAILS

Course Convener: Yang Shen

Course code: ACTL2102

Course name: Foundations of Actuarial Models

Name of this Assignment Item: Assignment 1

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STUDENT DETAILS

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Executive Summary

This report evaluates two ‘No Claim Discount’ (NCD) schemes for car insurance based on their short- and long-term proportions for discount levels. Without the NCD, the ten-year transition matrices reveal scheme A has an even distribution across six discount levels offered, whereas scheme B skews heavily towards the lowest discount level of the 4 available. The long-run average discounts were 12.30% for Scheme A and 14.83% for Scheme B. With NCD, the limiting probabilities for scheme A become more negatively skewed and more spread out for scheme B. With NCD, Scheme A's expected cost is \$673.97, and \$788.64 for B to insurers. With the NCD, insurers can expect to save \$749.06 per annum per policyholder, which results in net savings of \$75.10 and -\$39.57 for schemes A and B respectively, this makes scheme A better from insurers’ perspectives.

Part 1

1.1 – Transition Matrices for Schemes A and B

With the provided discounts for schemes A and B and the rate for the Poisson distribution, $\lambda = 0.7$ as seen in the background information provided. The probability of no claim is $e^{-0.7} = 0.4966$ and $1 - e^{-0.7} = 0.5034$ is the probability of there being at least one claim, these probabilities are consistent over all transitions, making them time homogenous. We found the transition matrices which represent the probability of transition from one state to another or remaining in the same state. The states correspond to the discount levels that policyholders are entitled to depending on whether they make a claim in a certain year or not and which scheme they choose. The definitions of these states are available in the background provided and Appendix 1.

$$P_A = \begin{pmatrix} 0.5034 & 0.4966 & 0 & 0 & 0 & 0 \\ 0.5034 & 0 & 0.4966 & 0 & 0 & 0 \\ 0 & 0.5034 & 0 & 0.4966 & 0 & 0 \\ 0 & 0 & 0.5034 & 0 & 0.4966 & 0 \\ 0 & 0 & 0 & 0.5034 & 0 & 0.4966 \\ 0 & 0 & 0 & 0 & 0.5034 & 0.4966 \end{pmatrix}$$
$$P_B = \begin{pmatrix} 0.5034 & 0.4966 & 0 & 0 \\ 0.5034 & 0 & 0.4966 & 0 \\ 0.5034 & 0 & 0 & 0.4966 \\ 0.5034 & 0 & 0 & 0.4966 \end{pmatrix}$$

This shows both schemes’ matrices are irreducible since all states are accessible from one another.

1.2 – Ten Year Transition Matrices

To calculate the ten-year transition matrix, we used R. We used the transition matrices P_A and P_B . We set the number of years we were computing the matrix for, which is ten years, meaning $n = 10$. We then used matrix exponentiation, essentially a series of matrix multiplication to compute the transition matrices over ten years.

$$P_A^{(10)} = \begin{pmatrix} 0.2513 & 0.2084 & 0.2055 & 0.1258 & 0.1241 & 0.0849 \\ 0.2112 & 0.2484 & 0.1275 & 0.2038 & 0.0861 & 0.1229 \\ 0.2112 & 0.1293 & 0.2466 & 0.0873 & 0.2026 & 0.1229 \\ 0.1311 & 0.2095 & 0.0885 & 0.2454 & 0.1246 & 0.2009 \\ 0.1311 & 0.0897 & 0.2082 & 0.1263 & 0.2437 & 0.2009 \\ 0.0909 & 0.1298 & 0.1281 & 0.2065 & 0.2037 & 0.2410 \end{pmatrix}$$

Policyholders will receive the discount that they started with for the largest proportion of time in ten years.

$$P_B^{(10)} = \begin{pmatrix} 0.5034 & 0.2500 & 0.1241 & 0.1225 \\ 0.5034 & 0.2500 & 0.1241 & 0.1225 \\ 0.5034 & 0.2500 & 0.1241 & 0.1225 \\ 0.5034 & 0.2500 & 0.1241 & 0.1225 \end{pmatrix}$$

Each row of the matrix is identical, indicating that the probability of transitioning from one state to another is the same, regardless of the current state. Policyholders can expect a discount of 0% for 50.34% of the ten years and a discount of 15% for a quarter of the ten years, they can expect to spend less time with higher discounts.

1.3 – Limiting Probabilities

To find the limiting probabilities, we solve a system of linear equations derived from the transition matrix, this will provide the long-run proportion of time spent in each state. The equations are found through $\pi P = \pi$ where π represents the limiting probabilities. These limiting probabilities remain unchanged after any time step. Further, these probabilities must sum to one. This provides a system of $n+1$ linear equations with $n+1$ unknown limiting probabilities, with $n=6$ in scheme A and $n=4$ in scheme B. We converted this system of equations into matrix form, created a function in R, and substituted transition matrices to output limiting probabilities.

$$\pi_A = (0.1724 \quad 0.1701 \quad 0.1678 \quad 0.1655 \quad 0.1632 \quad 0.1610)$$

$$\pi_B = (0.5034 \quad 0.2500 \quad 0.1241 \quad 0.1225)$$

The contrasting limiting probabilities for Scheme A and B reveal significant differences in their long-term discount distribution strategies. Scheme A's probabilities are more evenly spread across its six discount levels. All the discount level categories have a long-term probability of 16-17% which suggests policyholders spent approximately equal time in each state in the long run. In contrast, Scheme B exhibits a more positively skewed distribution with a dominant proportion at the lowest discount level of 0% and the proportions progressively decrease as the discount level increases, which means policyholders in B spend little time in higher discount states. Understanding these distributions is crucial for insurers to design insurance schemes that balance attracting and retaining customers with minimising costs.

1.4 – Average Discounts in Long Term

To find the average long-term discounts, through the limiting probabilities computed, we also defined the discount levels in R. We then found the average discount which was found by multiplying the limiting probability by the corresponding discount level and summing all products, this is the weighted discount level.

In the long run, the weighted discount levels portray the average discount level for scheme A as 12.30% and that of scheme B as 14.83%. Although there is only a 2.53% difference, this is significant since in the context of insurance it could represent a significant difference in cost to the policyholder and insurer.

1.5 – Simulating Markov Chain for Scheme A

To simulate the Markov chain with a transition matrix under Scheme A we used Monte Carlo simulations. The current state determines the next state based on cumulative probabilities from the transition matrix which is due to the nature of Monte Carlo simulations in which the outcome of uncertain events like this is determined based on probability and random number sampling. The stochastic nature of the process means results will vary over different runs hence results will become more accurate through more simulations. We first defined the transition matrix P_A , which is used to model the simulation paths. Next, we set the initial state that corresponds to the discount level of 0%, time horizon $n = 10$ and the number of simulations, $m = 1000$ simulations. This setup requires a total of 10,000 random variables since each path requires 10 random variables, one for each year.

We then randomly generated uniform random variables which were to be used to determine transitions. We used antithetic variates in which the counterpart of each random number was used to create antithetic paths, this induces a negative correlation between paired variables so that their combined effect stabilises the estimate, which reduces variance in results. The uniform random variables u and $1 - u$ are combined into a single sequence, ensuring that each path is influenced equally by both the original and antithetic variates.

We initialised a matrix containing the simulated paths, this matrix had 1000 rows for the number of simulations and eleven columns for each time step including time 0. Each path is simulated by iterating over the time horizon and applying random variables to determine transitions based on cumulative transition probabilities.

We then generated a single path with the 'simulate_path' function we defined, which simulated transitions over n years using the transition matrix P_A using the pre-generated random variables. In this function, we start from the initial state and for each time step, the next state is based on cumulative transition probabilities derived from the current state and the random variable. Each path represents a distinct policyholder's discount level evolution over time. Then, we used another function 'simulate_paths' to generate m paths by invoking 'simulate_path' and systematically applying both original and antithetic variates across the paths to balance randomness and reduce variance. We stored the results in a matrix where each row corresponds to a simulation path. The first few simulated paths were printed to confirm their validity. These simulated paths represent the possible variations of the evolution of discount levels for policyholders and account for the dynamic environment.

1.6 – Expected States of X_1 and X_{10}

We used derivations instead of simulations to find the expected states in one and ten years since there is no approximation involved and is more computationally straightforward since the calculations are based on exact mathematical relationships and Markov chain properties. We set the initial state vector as X_0 in R and multiplied this by the transition matrix P_A . This provided probabilities of transitioning into different states in one year. This process was repeated by multiplying the initial state vector by P_A^{10} to find the probabilities for ten years.

The state space is $\{e_1, e_2, e_3, e_4, e_5, e_6\}$, which corresponds to the progressive discount levels as provided in the background information and Appendix 1 for Scheme A. The transition probabilities given $X_0 = e_1$ are:

$$X_1 = (0.5034 \quad 0.4966 \quad 0 \quad 0 \quad 0 \quad 0) \\ X_{10} = (0.2513 \quad 0.2084 \quad 0.2055 \quad 0.1258 \quad 0.1241 \quad 0.0849)$$

In the vector, X_1 there is only one transition period, hence it is only possible to remain in the starting state or move up one, the expectation is that 50.34% of the time is spent in the initial state and 49.66% in state e_2 . X_{10} is more diversified due to ten transitions occurring. Progressively less time is spent in each subsequent state as the discount level increases, 25.13% of the time in ten years is spent with a discount of 0% and 8.49% at the highest level of 25%. These are just the expectations within ten years, definite statements cannot be made due to the dynamic nature of these exponentially distributed probabilities.

Part 2

2.1 – No Claim Discount (NCD) Transition Matrices and Limiting Probabilities

With an NCD, the transition matrixes will be adjusted to account for decreased claim frequency. This results from two effects, one being more careful driving which decreased the Poisson rate and the second being that the policyholder does not make a claim if the cost is less than \$1000. The initial claim frequency represented by $\lambda = 0.7$ indicates a relatively high claim frequency which is not ideal for us as insurers.

To find the transition matrices, we defined the average claim size (\$5000) and rate $\frac{1}{5000}$ in R. The claim frequency also decreased to $\lambda = 0.56$. Through the thinning Poisson process where we formed subgroups of over and under \$1000, we defined functions to represent the probabilities of at least one claim (36.78%) and no claims (63.22%). We found transition probabilities through exponential distribution.

$$P_{A(NCD)} = \begin{pmatrix} 0.3678 & 0.6322 & 0 & 0 & 0 & 0 \\ 0.3678 & 0 & 0.6322 & 0 & 0 & 0 \\ 0 & 0.3678 & 0 & 0.6322 & 0 & 0 \\ 0 & 0 & 0.3678 & 0 & 0.6322 & 0 \\ 0 & 0 & 0 & 0.3678 & 0 & 0.6322 \\ 0 & 0 & 0 & 0 & 0.3678 & 0.6322 \end{pmatrix} \\ P_{B(NCD)} = \begin{pmatrix} 0.3678 & 0.6322 & 0 & 0 \\ 0.3678 & 0 & 0.6322 & 0 \\ 0.3678 & 0 & 0 & 0.6322 \\ 0.3678 & 0 & 0 & 0.6322 \end{pmatrix}$$

To find the limiting probabilities, we applied the same method as we used to find P_A^{10} and P_B^{10} . We found –

$$\pi_A = (0.0290 \quad 0.0498 \quad 0.0856 \quad 0.1472 \quad 0.2531 \quad 0.4352) \\ \pi_B = (0.3678 \quad 0.2325 \quad 0.1470 \quad 0.2527)$$

This shows a drastic shift from the limiting probabilities without NCD, scheme A is significantly negatively skewed with NCD with the majority of the proportion of time being spent at a discount level of 25%. Scheme B is relatively less negatively skewed than it was without the NCD, although the highest proportion is still at a 0% discount level, there is more time being spent at higher discount levels, and policyholders expect to spend 25.27% of the time at a discount level of 55%. This means policyholders are rewarded for making fewer claims since they are more likely to move up discount levels and both schemes are now more appealing to policyholders.

2.2 – Expected Long Run Costs to Insurers with NCD

To compute the average discount, we multiplied each discount level by its corresponding limiting probability found in section 2.1 and summed it together. We found the raw premium without NCD to be the claim frequency (0.7) multiplied by the average claim size (\$5000), which resulted in \$3500. We then found the long-run average premium which was the *raw premium* * (1 – *average discount rate*). The average cost for each scheme was found by subtracting the long-run average premium from the raw premium.

We found the average long-run discount that policyholders would expect to receive with the NCD for schemes A and B is 19.26% and 22.53% respectively. This rate suggests that scheme B would be more appealing to policyholders due to the higher discount offered. Then, it was found that the expected cost per annum for insurers per policyholder for schemes A and B is \$ 673.97 and \$788.64 respectively. With the NCD, scheme B costs \$114.67 more than Scheme A for insurers without considering their revenue from implementing the NCD.

2.3 – Expected Savings on Insurance Claims with NCD

The expected savings in both schemes are independent of the discount levels. Hence, we expect the same savings in both schemes, as it is the expected amount a policyholder would theoretically claim which is consistent across schemes even with different discounts. We defined the average claim size and rate in R, to ensure only claims above \$1000 are included and the claim frequency is reduced based on reduced claim frequency from careful driving. We multiplied 0.56 (claim frequency after effect i) by the exponential distribution with rate $\frac{1}{5000}$ which is multiplied by 1000 for the claim size boundary. We used this since the time spent in each state follows an exponential distribution under the Poisson Process and is memoryless. This provides the adjusted claim frequency of 0.4585. We calculated the expected savings with NCD for both schemes by computing the expected aggregate claims using the original claim frequency and mean claim and subtracting the expected claim size with NCD effects. The expected claim size with effects is found by multiplying the adjusted claim frequency 0.4585 by 6000. This is because we are only considering claims greater than 1000 which means the average claim size is 5000+1000 where 5000 is the original mean claim size, this is explained in Appendix 10.

We found the expected savings for the company to be \$749.06 resulting from less expected claims, without considering the downside of the lower premium income due to discounts. This is beneficial for policyholders as they receive discounts, and supports our insurance strategic objectives; by managing claim costs and frequency, we can increase profitability from the anticipated higher demand for services, hence maintain competitiveness.

2.4 – Net Expected Savings Per Policyholder Per Annum

The net savings are found by finding the difference between the expected savings by policyholders and the expected costs involved for us as insurers. We found the net savings for schemes A and B to be \$75.10 and -\$39.57 respectively. Scheme A provides higher net savings and, hence is better from the insurer's perspective as it provides a higher profit after providing discounts to policyholders, which means we save \$75.10 per policyholder per annum. As seen in section 2.2, scheme A also provides policyholders with significant discounts, so it is a win-win situation. Scheme A is more cost-effective in the long run as it more efficiently reduces expected claim costs relative to the expenses. Scheme B should not be implemented as it would have an adverse effect on the company's standing as they would be making a net loss per policyholder. Scheme A's higher net savings contribute to insurers' profit margin which ensures financial stability and opportunities to invest in other strategic areas. Therefore, this NCD scheme can lead to higher profitability, more competitive pricing and financial stability through the higher anticipated policyholders due to discounts. The NCD provides more than just monetary benefits in the long run, it also ensures that as insurers we can better categorise high and low-risk policyholders and in turn further modify schemes to be able to offer tailored insurance products that meet the specific needs of different policyholders, enhancing customer satisfaction and retention.

Part 3

3.1 – Recommendations to Enhance Models: Addressing Current Limitations

As insurers, we want to maintain a profitable yet sustainable and customer-centric business. To improve schemes A and B, we recommend differentiating claims by fault. This would ensure fairness and that policyholders are not made to suffer due to situations where they are not at fault. However, this could come at a cost to insurance companies, hence there should be an associated penalty such as the forgoing of increased discount level in future year, whilst preventing policyholders from going down. This is to incentivise loyalty to the company even when a claim is made. This would be further improved through forgiveness policies where the insurer could forgive the first at-fault claim if the claim size were below a certain threshold which would involve a non-homogeneous Poisson process since it requires information regarding previous claims, this would help retain customers as they can have a higher interarrival time at higher discount levels. To further improve customer retention, we should consider policyholder loyalty and increase discounts or forgive the next claim for loyal policyholders who have for instance been loyal for ten years and made under three claims or an equivalent strategy, this would involve a non-homogeneous Poisson Process. We recommend more attention to the claim size distribution, larger claims might result in a greater reduction in the discount level compared to smaller claims. This would prevent minor claims from affecting the policyholder's status unreasonably and will likely retain customers since they find more utility due to more flexibility since the stakes are not as high and they feel fairly make claims instead of refraining due to fears of potential lower discounts in future. Customer retention is important as it can stabilise insurers' revenue and provide customers with a sense of security as become familiar with their scheme. These factors are analytically tractable as they can adjust to fit a Markov Chain by introducing non-homogeneous Poisson Processes, thus we can determine if the net savings remain positive after such implementations.

Appendix

Appendix 1 – Outline of the schemes

- Scheme (A) offers discounts of 0%, 5%, 10%, 15%, 20% and 25% on the premium. If no claim is made during the current year, then the policyholder will move up one discount level or remain at 25% in the next year. If one or more claims are made, then the policyholder will move down one level, or remain at 0% discount in the next year.
- Scheme (B) offers discounts of 0%, 15%, 35% and 55% on the premium. If no claim is made during the current year, then the policyholder will move up one discount level or remain at 55% in the next year. If one or more claims are made, then the policyholder will move back to 0% discount, i.e. the NCD is forfeited, in the next year.
- Under both Schemes (A) and (B), all new policyholders start at a discount level 0%.

Appendix 2 – Part 1 Question 1

$$P_A = \begin{pmatrix} P(\text{claim}) & P(\text{no claim}) & 0 & 0 & 0 & 0 \\ P(\text{claim}) & 0 & P(\text{no claim}) & 0 & 0 & 0 \\ 0 & P(\text{claim}) & 0 & P(\text{no claim}) & 0 & 0 \\ 0 & 0 & P(\text{claim}) & 0 & P(\text{no claim}) & 0 \\ 0 & 0 & 0 & P(\text{claim}) & 0 & P(\text{no claim}) \\ 0 & 0 & 0 & 0 & P(\text{claim}) & P(\text{no claim}) \end{pmatrix}$$

$$P_B = \begin{pmatrix} P(\text{claim}) & P(\text{no claim}) & 0 & 0 \\ P(\text{claim}) & 0 & P(\text{no claim}) & 0 \\ P(\text{claim}) & 0 & 0 & P(\text{no claim}) \\ P(\text{claim}) & 0 & 0 & P(\text{no claim}) \end{pmatrix}$$

We understood that we had to apply all assertions given in the background information regarding transitions. Hence we defined that the probabilities of going from state i to j had a specific pattern as illustrated above.

Appendix 3 – Part 1 Question 2

To find the 10 year transition matrices, we applied matrix exponentiation

$$P_A^{10} = \begin{pmatrix} 0.5034 & 0.4966 & 0 & 0 & 0 & 0 \\ 0.5034 & 0 & 0.4966 & 0 & 0 & 0 \\ 0 & 0.5034 & 0 & 0.4966 & 0 & 0 \\ 0 & 0 & 0.5034 & 0 & 0.4966 & 0 \\ 0 & 0 & 0 & 0.5034 & 0 & 0.4966 \\ 0 & 0 & 0 & 0 & 0.5034 & 0.4966 \end{pmatrix}^{10}$$

$$P_B^{10} = \begin{pmatrix} 0.5034 & 0.4966 & 0 & 0 \\ 0.5034 & 0 & 0.4966 & 0 \\ 0.5034 & 0 & 0 & 0.4966 \\ 0.5034 & 0 & 0 & 0.4966 \end{pmatrix}^{10}$$

Appendix 4 – Part 1 Question 3

To find limiting probabilities, we matrix multiplied the limiting probabilities vector denoted by π_i 's by the transition matrix.

Our system of equations were found to be

Scheme A

$$\begin{cases} \pi_1(0.5034) + \pi_2(0.5034) - \pi_1 = 0 \\ \pi_1(0.4966) + \pi_3(0.5034) - \pi_2 = 0 \\ \pi_2(0.4966) + \pi_4(0.5034) - \pi_3 = 0 \\ \pi_3(0.4966) + \pi_5(0.5034) - \pi_4 = 0 \\ \pi_4(0.4966) + \pi_6(0.5034) - \pi_5 = 0 \\ \pi_5(0.4966) + \pi_6(0.4966) - \pi_6 = 0 \\ \pi_1 + \pi_2 + \pi_3 + \pi_4 + \pi_5 + \pi_6 = 1 \end{cases}$$

This system of equations was solved in R by converting it to matrix form.

Scheme B

$$\begin{cases} \pi_1(0.5034) + \pi_2(0.5034) + \pi_3(0.5034) + \pi_4(0.5034) - \pi_1 = 0 \\ \pi_1(0.4966) - \pi_2 = 0 \\ \pi_2(0.4966) - \pi_3 = 0 \\ \pi_3(0.4966) + \pi_3(0.4966) - \pi_4 = 0 \\ \pi_1 + \pi_2 + \pi_3 + \pi_4 = 1 \end{cases}$$

This system of equations was solved in R by converting it to matrix form.

Appendix 5 – Part 1 Question 4

The weighted average discount level in the long term is simply the sum of all long-run proportions of time to see how much time is spent in each state in the long run, multiplied by the discount level in that state. The sum of this gives the weighted average discount.

Appendix 6 –Method for simulating the Markov chain with transition matrix under Scheme (A) from time 0 to time n

```
m <- 1000 # Number of simulations
n <- 10   # Time horizon
```

```
# Function to simulate a single path of the Markov chain using specified random variables
```

```
simulate_path <- function(T, n, us, index) {
  path <- numeric(n + 1) # Initialise path vector with n+1 elements
  path[1] <- 1 # Start at discount level 0 (index 1)
  for (i in 1:n) {
    u <- us[index + i - 1] # Use the provided random variable from the pre-generated list
    current_state <- path[i] # Get the current state from the path
    transition_probs <- T[current_state, ] # Get transition probabilities for the current state
    path[i + 1] <- which.max(cumsum(transition_probs) >= u) # Determine the next state
  }
  return(path - 1) # Convert to discount level (adjust indexing)
}
```

```
# Function to simulate multiple paths using pre-generated random variables and antithetic variates
```

```
simulate_paths <- function(T, n, m) {
  paths <- matrix(0, nrow = m, ncol = n + 1) # Initialise matrix to store all paths
  us <- runif(m * n / 2) # Generate half of the required uniform random variables
  antithetic_us <- 1 - us # Generate the antithetic variates
  us <- c(us, antithetic_us) # Combine the original and antithetic variates
  for (i in 1:m) {
    start_index <- ((i - 1) * n) %% length(us) + 1 # Calculate the starting index for the current path
    paths[i, ] <- simulate_path(T, n, us, start_index) # Simulate path using the random variables
  }
  return(paths)
}
```

```
# Perform the simulation for transition matrix A with given parameters
```



```
paths <- simulate_paths(T_A, n, m)
```

```
# Print the first few paths to confirm if results are as expected
print(head(paths))
```

Note
500 paths with 5000 uniform random variables path and 500 with 5000 antithetic variables which is with the complement of the uniform random variable.

Appendix 7 – Part 1 Question 6

For discrete time Markov chains

$$E(X(n)|X(0) = e_i) = e_i * P$$

Where P is a transition matrix and e_i is the starting vector which in our case is $e_i = X_0$.

Appendix 8 – Part 2 Question 7

Mean Claim size: $\mu = 5000$

Claim Rate: $\lambda = \frac{1}{\mu} = \frac{1}{5000}$

The time spent in each state follows an exponential distribution. The transitions also follow a Poisson process since it is a model which is a series of discrete time events and the average time between the events is known however the timing of the events is random which follows an exponential distribution and that in itself is memoryless. This random variable is known as the interarrival time (waiting time between events is memoryless) and it strictly follows exponential distribution in the Poisson process.

Appendix 9 – Part 2 Question 8

The average discount is computed with the same method as appendix 5.

$$\text{Raw Premium} = \text{Claim Frequency} * \text{Mean Claim size} = 0.7 * 5000 = 3500$$

$$\text{Long Run Average Premium} = \text{Raw Premium} * (1 - \text{Average discount rate for each scheme})$$

$$\text{Expected Cost to Insurer} = \text{Raw Premium} - \text{Long Run Average Premium}$$

Appendix 10 – Part 2 Question 9

Adjusted claim frequency:

$$\lambda = 0.56 * e^{\frac{-1000}{5000}}$$

The reason for using \$6000 instead of 5000 to calculate the expected claim size

The average claim size is \$5000, which means that the claim rate is $\frac{1}{5000}$.

$$\begin{aligned} E[\text{Claim size} | \text{Claim size} > 1000] &= 1000 + \frac{1}{\text{claim rate}} \\ &= 1000 + 5000 \\ &= 6000 \end{aligned}$$

We are only considering claims greater than 1000 which means the claim size we consider is 1000+5000 (since 5000 is the mean without NCD) so minor claims are not included, minor being less than 1000 over the mean. Hence we use this rate and multiply by the modified lambda to find the expected savings.

$$\text{Expected mean claim size} = 6000 * \text{Adjusted Claim Frequency}$$

$$\text{Expected savings} = \text{Aggregate claims without effects} - \text{Aggregate claims with effects}$$

Appendix 11 – Part 2 Question 10

$$\text{Net Savings} = \text{Expected savings for policyholder} - \text{Expected cost to insurer}$$

Generative AI Appendix

- How to compute the expected costs of running insurance schemes?
- How to calculate expected savings on insurance claims?
- How to find transition matrices and limiting probabilities for different schemes?
- Are the following transition matrices I provided constructed in the correct format for the Markov chain?
- How to define claim frequency using a Poisson process?
- When to use ``dpois`` vs. ``pexp`` for probability calculations?
- How do I include the threshold into the Poisson process rates?
- What are the best methods to simulate a Markov chain using transition matrices?
- Should I use antithetic variates for variance reduction in simulations and what do they do?
- How to compare net expected savings between different insurance schemes?
- Which scheme is better from the insurer's perspective based on expected costs and savings?
- What is the difference between ``exp`` and ``pexp``?
- How should I use the derivations method to find the state in a certain amount of time?
- How do I simulate Markov chains and include antithetic variables in it?
- How can I do what is in the 'steady state' function manually?
- How do I use the Markov chain package in R?
- How do I use the `expm` package in R?
- How do I use the `'cat'` function in R?
- How do I lay out a matrix in R?
- How do I do matrix exponentiation in R?
- What are the different ways I can find limiting probabilities without using the Markov chain package, preferably using systems of equations?
- How do I do simulations with antithetic variates in R?